Name & Surname **Polimi ID**

Exercise 1

A. In which way does the following assumption impact the PRAM analysis: "Each memory cell can hold an integer of unbounded size"? (1)

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B. Could you please classify the READ/WRITE abilities of a PRAM system and how realistic and useful they are? (1)

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C. Assuming T*(n) = T1(n), what kind of relation do you have between the efficiency and the speedup of a PRAM-based application (direct, inverse,…)? (1)

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D. Could you please define the speedup in a fixed-time model? (1)

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Exercise 2

A. Show an example with a few lines of code where a data-parallel program allows SIMD/vectorize instructions extraction. (1)

B. Please describe what stream coherence means and why it is related to SIMD processing resources. (2)

C. Write a short description of "Hardware-supported interleaved multi-threading." (1)

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Exercise 3

A. Please describe the thread hierarchy in the CUDA abstraction. Is a “warp” part of the CUDA language? (1)

B. Please describe the main characteristics of a message-passing programming model. (1)

C. Multi-threading is mainly meant for **throughput-oriented systems**. True/False. Why? (2)

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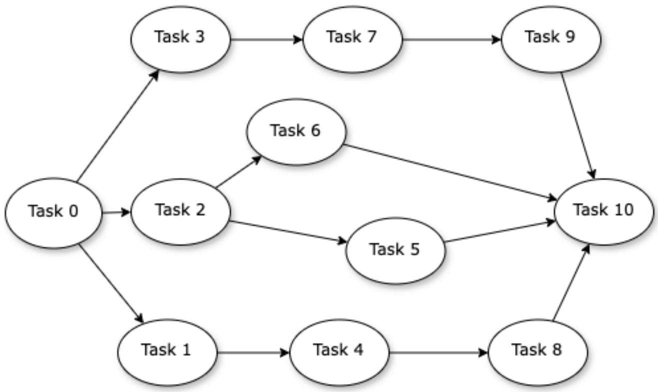
Exercise 4

A. Which methods are available in OpenMP to protect access to a shared variable? (1)

Consider the following task graph:



W(Task 0) = 1
W(Task 1) = 100
W(Task 2) = 100
W(Task 3) = 100
W(Task 4) = 100
W(Task 5) = 100
W(Task 6) = 100
W(Task 7) = 100
W(Task 8) = 100
W(Task 9) = 100
W(Task 10) = 1

B. Is it a good idea to implement the program with 4 threads? What is the theoretical improvement in execution time with respect to a sequential version? (1) **This question can be skipped if you got 1 or 3 points in the first challenge**

C. Write a corresponding OpenMP implementation. (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**